

OA Test Day 01: **R. Sai Sri Laasya**

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Write your Name and Registration no as well ans submit it on Whatsapp itself in pdf format use only pen and paper and no online/chatgpt help

I am only taking this to check which part you should focus more:

NO Grading : so try to solve it by yourself. I may ask how you have done it. To anyone

Time is till 12:45....

Best Of Luck

? Next number in 3, 7, 15, 31, → 63

? If $8x = 56$, find x → 7

? Average of 5 numbers is 20; sum of first 3 is 42; find sum of last 2 → 58

? Profit % when CP = ₹200 and SP = ₹250 → 25

? 10% of 450 → 45

? Time-speed-distance: 60 km in 1.5 hrs → 11m/s

? LCM of 12 and 18 → 36

? 15% of 360 → 54

? Reduce 75 by 20% → 60

? Simple interest on ₹8000 at 10% p.a. for 2 years → 1600

? Ratio 7:9 of 160 → 70 and 90

? $(5x + 3) = 23$ solve for x → 4

? Perimeter of rectangle 30 m, width 5 m , length → 10

? Area of square with side 7 → 49

- ❓ Volume of cube with edge 3 → 27
- ❓ % increase from 80 to 100 → 25
- ❓ Find the median of 4, 9, 15, 20 → 12
- ❓ Find the mode of 2,2,3,4,4,4 → 4
- ❓ Compound interest for 2 yrs
- ❓ Time and work: A and B together finish in 6 days
- ❓ Probability of rolling even number on dice → $\frac{1}{2}$
- ❓ Find 3 consecutive even integers sum 48 → 14, 16, 18
- ❓ Discount of 15% on ₹1200 → discount = 180 and sp = 1020
- ❓ Distance between (3,4) and (6,8) → 5
- ❓ A train 100m enters tunnel 200m at 60 km/hr → 18 sec
- ❓ Mixture ratio 3:2 total 50 liters → 30 and 20 litres
- ❓ Age problem: Father is 3x son's age
- ❓ Calendar: Day after Tuesday → wednesday
- ❓ Circular track 400m, 5 laps distance → 2000
- ❓ Simple equation: $3x - 5 = 16$ → $x = 7$

Logical Reasoning (30 Questions)

Number & Letter Series

- 31. 2, 6, 12, 20, ___ → 30
- 32. 5, 11, 23, 47, ___ → 95
- 33. V, X, Z, ___ → B

Analogies & Classification

- 34. Cat: Kitten :: Dog: ___ → puppy
- 35. Which doesn't belong: Apple, Orange, Carrot, Mango → carrot
- 36. Pigeon: Bird :: Shark: ___ → fish

Blood Relations

37. A is father of B; B is sister of C — relationship C to A? → C is child of father A
38. If M is uncle of N and P is father of M, what is P to N? → P is grand father of N

Syllogisms

- 39–40. Standard 2-statement/ conclusion patterns

Direction Sense

41. If you face east, turn right twice — final direction? → west
42. Walk 5m N, 3m E, 5m S – where relative? → 3m east

Seating Arrangements

- 43–44. Linear arrangement True/False problems

Puzzles & Critical Reasoning

45. Logical puzzle about 5 people heights
46. If all cats are pets and some pets are big animals – which follows?

Visual/Pattern Reasoning

- 47–50. Shape sequence pattern
51–52. Mirror & water image reasoning

Verbal Logical Reasoning

53. Statement: “All roses are flowers; all flowers fade” does it imply “All roses fade”? → Yes
54. If $A \Rightarrow \neg B$ and $B \Rightarrow C$, can A be true with C true? → No



Verbal Ability (30 Questions)

Grammar & Usage

55. Identify the verb: “He ___ the book.” (choose correct) → reads
56. Correct sentence: (a) Him and me went... (b) He and I went... → B is correct
57. Fill blank with preposition

Synonyms/Antonyms

58. Synonym of “Happy”
59. Antonym of “Scarce”
60. Choose closest meaning to “Eloquent”

Reading Comprehension

- 61–63. Short passage questions (main idea / inference)

Sentence Completion

64–66. Fill blanks in sentence meaningfully

Error Spotting

67–70. Identify error in sentences

Para-jumbles

71–74. Arrange jumbled sentences to form coherent paragraph

Odd One Out

75–78. Choose word not belonging to group

Idioms & Phrases

79–82. Meaning of commonly used idioms

One-word Substitutes

83–86. e.g., “One who loves books”

87–90. e.g., “Fear of heights”

Computer Basics (1–10)

1. What is a computer?

- A. An electronic toy
- B. A machine that performs user-given operations**
- C. A storage device
- D. Only an input device

2. Which component is known as the brain of the computer?

- A. RAM
- B. Hard Disk

C. CPU

- D. Monitor

3. CPU mainly consists of which two units?

- A. RAM & ROM
- B. ALU & CU**
- C. Cache & Register
- D. Input & Output

4. Which memory is volatile in nature?

- A. Hard Disk
- B. ROM

C. RAM

- D. Pen Drive

5. Which memory is also called Primary or Working Memory?

- A. Hard Disk
- B. RAM**
- C. Cache
- D. ROM

6. CPU can directly access which memory?

- A. Hard Disk
- B. Pen Drive
- C. Main Memory (RAM)**
- D. DVD

7. Which device is slower but can store large amounts of data?

- A. Cache
- B. RAM

C. Hard Disk

D. Register

8. Which of the following is an input device?

A. Monitor

B. Printer

C. Keyboard

D. Speaker

9. Output devices have their own memory called:

A. Cache

B. Output Buffer

C. RAM

D. ROM

10. What is the correct flow of data processing?

A. CPU → Input → Output → Memory

B. Input → CPU → Output → Hard Disk

C. Input → Memory → CPU → Output

D. Hard Disk → CPU → Input → Output

Programming Languages & Translators (11–20)

11. What language does a computer understand?

A. English

B. C#

C. Binary

D. Java

12. Machine language consists of:

A. English words

B. Symbols

C. 0s and 1s

D. ASCII characters

13. Why is machine language difficult to use?

A. It is slow

B. It is error-prone and hard to understand

C. It needs compiler

D. It is portable

14. Assembly language is also called:

- A. High-level language
- B. Machine language
- C. Symbolic language**
- D. Scripting language

15. Which translator converts assembly code to machine code?

- A. Compiler
- B. Interpreter
- C. Assembler**
- D. Loader

16. Which translator converts high-level code all at once?

- A. Interpreter
- B. Compiler**
- C. Assembler
- D. Loader

17. Interpreter works in which manner?

- A. Line by line**
- B. All at once
- C. Binary only
- D. Register wise

18. Which language is compiler-based interpreted?

- A. C
- B. C++
- C. Java**
- D. Assembly

19. Source code is written in:

- A. Binary language
- B. Machine code
- C. High-level language**
- D. Assembly only

20. Object code is also called:

- A. Source code
- B. Machine code**
- C. Pseudocode
- D. Algorithm

Software & Operating System (21–28)

21. Software is defined as:

- A. Hardware devices
- B. Collection of programs**
- C. Memory unit
- D. CPU instructions only

22. Which is a system software?

- A. MS Word
- B. Tally

C. Operating System

- D. Browser

23. Which software gets auto-loaded when computer starts?

- A. Application software
- B. Compiler

C. Operating System

- D. Interpreter

24. Which program loads machine code into memory?

- A. Linker

B. Loader

- C. Compiler
- D. Editor

25. Which program combines multiple modules?

- A. Loader

B. Linker

- C. Interpreter
- D. Assembler

26. Which application needs installation on a machine?

- A. Web application

B. Standalone application

- C. Cloud app
- D. Mobile app

27. Gmail and YouTube are examples of:

- A. Standalone applications

B. Desktop applications

C. Web applications

D. Console applications

28. Programming languages are:

A. Web applications

B. Standalone applications

C. Mobile apps

D. Browser-based

Programming Methodologies (29–35)

29. Writing entire code in a single block is called:

A. Modular programming

B. OOP

C. Monolithic programming

D. Procedural programming

30. Biggest drawback of monolithic programming is:

A. Easy debugging

B. Code reusability

C. Difficult maintenance

D. Faster execution

31. Modular programming mainly improves:

A. Memory usage

B. Reusability and readability

C. Hardware usage

D. Compilation speed

32. OOP groups data and functions into:

A. Structures

B. Objects

C. Classes

D. Modules

33. Which is NOT an OOP feature?

A. Encapsulation

B. Inheritance

C. Compilation

D. Polymorphism

34. Which programming style supports teamwork easily?

A. Monolithic

B. Modular

C. Assembly

D. Machine-level

35. Reusability is best achieved using:

A. Monolithic programming

B. Assembly language

C. OOP

D. Machine language

.NET & C# Basics (36–45)

36. What does .NET provide?

A. CLR only

B. BCL only

C. CLR and BCL

D. Compiler only

37. CLR is responsible for:

A. Writing code

B. Converting MSIL to native code

C. Debugging only

D. Input handling

38. MSIL is generated during:

A. Execution

B. First compilation

C. Second compilation

D. Runtime only

39. JIT compiler works under:

A. BCL

B. OS

C. CLR

D. IDE

40. C# avoids memory leakage because of:

- A. Compiler
- B. Programmer

C. CLR garbage collection

- D. Interpreter

41. Visual Studio is:

- A. Compiler
- B. Language

C. IDE

- D. Framework

42. Entry point of a C# program is:

- A. Class
- B. Namespace

C. Main() method

- D. Object

43. Which keyword is used to include namespaces?

- A. include

B. using

- C. import
- D. namespace

44. Value types store data:

- A. By reference
- B. In heap

C. Directly in memory

- D. In files

45. int, char, bool in C# are:

- A. Reference types

B. Value types

- C. Pointer types
- D. Dynamic types

◆ **Type Casting & Conversion (1–12)**

1. Which type of conversion is done automatically by C#?

A. Explicit

B. Implicit

C. Boxing

D. Parsing

2. Which conversion may cause data loss?

A. int → long

B. float → double

C. double → int

D. short → int

3. What will be the output?

```
double d = 10.7;
```

```
int x = (int)d;
```

```
Console.WriteLine(x);
```

A. 11

B. 10

C. 10.7

D. Runtime error

4. Which method throws an exception if conversion fails?

A. TryParse

B. Convert.ToInt32

C. int.Parse

D. Both B and C

5. The return type of int.TryParse() is:

A. int

B. string

C. bool

D. void

6. What happens when int.Parse("abc") is executed?

A. Returns 0

B. Compile-time error

C. Runtime exception

D. Returns -1

7. Which conversion approach is safest?
- A. Explicit cast
 - B. `int.Parse`
 - C. `Convert.ToInt32`
 - D. `TryParse`**
8. Boxing in C# means:
- A. Value type to reference type**
 - B. Reference type to value type
 - C. string to int
 - D. double to int
9. Which operator is used for explicit casting?
- A. `as`
 - B. `is`
 - C. `()`**
 - D. `typeof`
10. What will be the output?

```
int a = 5;
```

```
double b = a;
```

```
Console.WriteLine(b);
```

- A. 5
- B. 5.0**
- C. Error
- D. Undefined

11. Which keyword is used to check type compatibility?
- A. `typeof`
 - B. `is`**
 - C. `as`
 - D. `new`
12. Unboxing converts:
- A. Reference to value**
 - B. Value to reference
 - C. Object to string
 - D. Double to int

◆ **Variables & Constants (13–22)**

13. Which keyword declares a constant variable?
- A. readonly
 - B. static
 - C. const**
 - D. final
14. A const variable must be initialized:
- A. At runtime
 - B. Inside constructor
 - C. At declaration**
 - D. Anywhere
15. Which variable has only one copy per class?
- A. Local
 - B. Instance
 - C. Static**
 - D. Dynamic
16. What is the default value of an instance int variable?
- A. null
 - B. 0**
 - C. undefined
 - D. garbage
17. Which variable is stored in stack memory?
- A. Static
 - B. Instance
 - C. Local**
 - D. Heap variable
18. Which modifier allows a variable to be assigned only once?
- A. static
 - B. const
 - C. readonly**
 - D. volatile
19. Can a const variable be static?
- A. Yes**

- B. No
- C. Only in structs
- D. Only in classes

20. Which variable is shared across all objects of a class?

- A. Instance
- B. Local
- C. Static**
- D. Readonly

21. Which keyword prevents modification after constructor execution?

- A. const
- B. static
- C. readonly**
- D. sealed

22. Which variable is destroyed after method execution?

- A. Static
- B. Instance
- C. Local**
- D. Global

◆ **Static vs Non-Static (23–32)**

23. Static members belong to:

- A. Object
- B. Method
- C. Class**
- D. Namespace

24. Which method can be called without creating an object?

- A. Non-static
- B. Virtual
- C. Static**
- D. Abstract

25. Static methods can access:

- A. Only instance variables
- B. Only static variables**

- C. Both
- D. Neither

26. How many copies of static variable exist?

- A. One per object
- B. One per class**
- C. One per method
- D. Multiple

27. Can a static method access non-static members directly?

- A. Yes
- B. No**
- C. Sometimes
- D. Only using base

28. Static constructors are called:

- A. Every time object is created
- B. Only once**
- C. At program end
- D. Manually

29. Which keyword is mandatory for static members?

- A. new
- B. static**
- C. this
- D. base

30. Can a class have multiple static constructors?

- A. Yes
- B. No**
- C. Depends on CLR
- D. Depends on access modifier

31. Static constructor cannot have:

- A. Parameters
- B. Access modifier
- C. Return type
- D. All of the above**

32. Static variables are initialized:

- A. After object creation
- B. Before first use**

- C. After Main()
 - D. At program termination
-

◆ **Operators (33–40)**

33. Which operator is used for null checking?

- A. ??**
- B. ?:
- C. &&
- D. ||

34. Which operator has highest precedence?

- A. +
- B. ***
- C. =
- D. &&

35. Output?

```
int x = 5;
```

```
Console.WriteLine(x++);
```

- A. 4
- B. 5**
- C. 6
- D. Error

36. Which operator checks equality?

- A. =
- B. ==**
- C. !=
- D. <=

37. Logical AND operator is:

- A. &
- B. &&**
- C. ||
- D. !

38. Ternary operator symbol is

- A. ::

B. ??

C. ?:

D. =>

39. Which operator is used for bitwise OR?

A. ||

B. |

C. &

D. ^

40. What does ! operator do?

A. Assignment

B. Negation

C. Comparison

D. Increment

◆ Basics of Data Types (1–6)

1. What information does a data type specify in C#?

A. Only memory size

B. Only range of values

C. Size, range, operations, and result type

D. Only data format

2. Computers internally represent all data using:

A. Decimal numbers

B. ASCII characters

C. Binary digits

D. Hexadecimal numbers

3. How many bits make one byte?

A. 4

B. 8

C. 16

D. 32

4. The binary representation of character 'A' is based on:

A. Unicode

B. UTF-16

C. ASCII

D. UTF-8

5. Which of the following is the smallest unit of data storage?

A. Byte

B. Bit

C. Word

D. Register

6. Which format is easier for humans to use while programming?

A. Binary

B. Hexadecimal

C. Decimal

D. Octal

◆ **Byte & SByte (7–12)**

7. What is the size of byte data type in C#?

A. 4 bits

B. 8 bits

C. 16 bits

D. 32 bits

8. The range of byte data type is:

A. -128 to 127

B. 0 to 127

C. 0 to 255

D. -255 to 255

9. Which data type allows negative values?

A. byte

B. sbyte

C. char

D. bool

10. How many total values can an 8-bit unsigned integer store?

A. 128

B. 255

C. 256

D. 512

11. Which data type should be used to store ASCII values only?

A. char

B. byte

C. string

D. int

12. What happens if you assign 256 to a byte variable?

A. Runtime error

B. Compilation error

C. Value becomes 0

D. Implicit conversion

◆ **Char & String (13–18)**

13. What is the size of char data type in C#?

A. 1 byte

B. 2 bytes

C. 4 bytes

D. 8 bytes

14. Char data type internally uses:

A. ASCII

B. UTF-8

C. UTF-16

D. UTF-32

15. Which data type can store regional language characters?

A. byte

B. char

C. bool

D. int

16. Which of the following is true about string in C#?

A. Value type

B. Pointer type

C. Reference type

D. Primitive type

17. What happens if you assign multiple characters to a char variable?

- A. Runtime exception
- B. Compilation error**
- C. Logical error
- D. Implicit conversion

18. A string is internally a collection of:

- A. bytes
- B. integers
- C. chars**
- D. objects

◆ **Integer Numeric Types (19–26)**

19. Which data type represents a 16-bit signed integer?

- A. int
- B. short**
- C. long
- D. byte

20. What is the maximum value of Int16?

- A. 32768
- B. 32767**
- C. 65535
- D. 2147483647

21. Which keyword is an alias for Int32?

- A. short
- B. int**
- C. long
- D. byte

22. Which data type should be used to store only positive numbers with larger range?

- A. short
- B. ushort**
- C. int
- D. sbyte

23. What is the size of long in C#?

- A. 4 bytes

B. 6 bytes

C. 8 bytes

D. 16 bytes

24. Which data type is unsigned?

A. int

B. short

C. uint

D. sbyte

25. Which data type stores the largest range of integer values?

A. int

B. short

C. long

D. byte

26. Which of the following supports only positive values?

A. sbyte

B. int

C. ushort

D. short

◆ **Floating-Point & Decimal (27–32)**

27. Which data type provides the highest precision?

A. float

B. double

C. decimal

D. long

28. Which suffix is required for decimal literals?

A. F

B. D

C. M

D. L

29. What is the size of float data type?

A. 2 bytes

B. 4 bytes

- C. 8 bytes
- D. 16 bytes

30. Which data type is best suited for financial calculations?

- A. float
- B. double
- C. decimal**
- D. int

31. Which floating-point type is least accurate?

- A. float**
- B. double
- C. decimal
- D. long

32. Which data type consumes more memory and is slower in performance?

- A. short
- B. int
- C. float
- D. decimal**

◆ **Reference, Pointer & Literals (33–40)**

33. Which of the following is a reference data type?

- A. int
- B. char
- C. string**
- D. byte

34. Which operator is used to get the address of a variable?

- A. *
- B. &**
- C. #
- D. %

35. Pointer usage in C# requires:

- A. checked block
- B. try-catch block
- C. unsafe keyword**
- D. static keyword

36. What does * operator represent in pointer programming?
- A. Address operator
 - B. Indirection operator**
 - C. Logical AND
 - D. Multiplication
37. Fixed values assigned to variables are called:
- A. Constants
 - B. Literals**
 - C. Keywords
 - D. Identifiers
38. Which prefix is used for binary literals in C#?
- A. 0x
 - B. 0b**
 - C. 0d
 - D. 0h
39. Hexadecimal literals in C# use base:
- A. 2
 - B. 8
 - C. 10
 - D. 16**
40. Which method safely converts string to integer without throwing exception? A.
- Parse()
 - B. Convert.ToInt32()
 - C. TryParse()**
 - D. Cast operator

<https://www.codechef.com/problems/CHEFEZQ>

<https://www.codechef.com/problems/SUBGCD>